Artifact Appendix for StickShield: Flexible Acoustic Metamaterial Defense for Securing Microphones Against Voice Attacks

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A Artifact Appendix

A.1 Abstract

StickShield is a flexible acoustic metamaterial patch that can be directly attached to the surface of a microphone. It is designed to defend against adversarial audio attacks, inaudible ultrasonic attacks, and laser injection attacks by absorbing acoustic energy within specific frequency bands while preserving the intelligibility of normal voice commands. This artifact includes 3D printing model files in `.SLDPRT` and `.STL` formats and demonstration videos in `.mp4` format. The `.SLDPRT`/`.STL` files are used to fabricate the metamaterial units, and the `.mp4` videos showcase StickShield's defensive effectiveness against typical inaudible and adversarial attacks. The artifact does not rely on any dataset. The artifacts are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).

A.2 Description & Requirements

The StickShield unit consists of two structural layers. The top layer is a membrane made of silicone rubber, with a Young's modulus of 9×10^6 Pa and a density of 1200 kg/m^3, primarily used to reflect and absorb high-frequency ultrasonic signals. The bottom layer is a box-like cavity structure made of soft polyurethane rubber, with a Young's modulus of 6.5×10^6 Pa and a density of 1040 kg/m^3, used to absorb adversarial perturbations in the audible frequency band.

The `.SLDPRT` files require SolidWorks to open and allow users to adjust the unit's geometric parameters according to the actual microphone dimensions of the target device. The `.STL` files are in standard 3D printing format and can be directly used to print individual units. Multiple units can be combined via copy and paste to accommodate different microphone array layouts. After printing, the structure is attached to the device's microphone surface with the top side facing outward, using double-sided tape or other adhesives or fastening mechanisms.

General artifact organization:

- `README.md`: overview of materials, fabrication, attachment, and the file list.
- `StickShield.SLDPRT`: editable SolidWorks model for geometric adjustment.
- `StickShield.STL`: printable model for fabrication.
- `StickShield-adversarial attack.mp4`: demonstration video for typical adversarial audio attacks.
- `StickShield-inaudible attack.mp4`: demonstration video for typical inaudible ultrasonic attacks.

The runtime environment consists of 3D printing equipment and video players for viewing demonstrations. An SLA 3D printer with a precision of at least 0.1 mm is recommended. The overall experimental workflow includes printing the unit, attaching it to the microphone, launching the attack, and recording the defense outcome. Printing takes about 2–4 hours, and attachment and testing take about 30 minutes.

A.2.1 Security, privacy, and ethical concerns. This work is intended solely for defensive security research and does not involve the collection of sensitive data from human subjects. Attack tests inject synthetic voice commands into a microphone or voice assistant and may cause the target device to execute commands. Users should avoid running attack tests on production devices, devices linked to sensitive accounts, or devices containing private data. If laser injection experiments are performed, users must follow appropriate laser safety procedures and avoid eye or skin exposure. No destructive steps or disabling of security mechanisms is required by the artifact itself.

A.2.2 How to access. The artifact is publicly available on GitHub at https://github.com/helo0507/StickShield. For the Artifacts Available badge, the permanently archived copy is available at https://doi.org/10.5281/zenodo.23874827. The GitHub repository and the Zenodo record refer to the same artifact.

A.2.3 Hardware dependencies.

- SLA or FDM 3D printer; an SLA printer with precision of at least 0.1 mm is recommended.
- Target microphone or device with a microphone/voice assistant.

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- Standard attack devices used to inject adversarial audio and inaudible ultrasonic commands.
- Video player for viewing `.mp4` demonstrations.
- Laser injection tests are not included in the current artifact package. If users independently replicate them, appropriate laser equipment and safety controls are required.

A.2.4 Software dependencies.

- SolidWorks to open and edit `.SLDPRT` files.
- Slicing software to generate G-code from `.STL` files.
- 3D printer firmware/control software.
- Standard video player for `.mp4` files.
- No source code compilation is required.

A.2.5 Benchmarks. None. The artifact does not rely on any dataset. Attack tests use standard synthetic voice commands as test inputs rather than a benchmark dataset.

A.3 Set Up

A.3.1 Installation.

(1) Obtain the 3D blueprints from the open-science platform.
(2) Open the `.SLDPRT` files in SolidWorks and adjust the unit's length, width, and height according to the target microphone size.
(3) Use slicing software to generate G-code.
(4) Print the part using an SLA or FDM 3D printer. The top layer is recommended to be printed in silicone rubber, and the bottom layer in soft polyurethane rubber.
(5) Remove support structures and clean the surface before attachment.
(6) Attach the printed structure to the device's microphone surface with the top side facing outward using double-sided tape or another adhesive/fastening mechanism.

A.3.2 Basic test. Open the `.STL` file in a standard STL viewer or SolidWorks and open the `.SLDPRT` file in SolidWorks. Verify that the model contains the two-layer StickShield geometry and that dimensions can be adjusted. Play the `.mp4` demonstration videos with a standard video player. Expected successful output: the geometry renders correctly, the editable model opens in SolidWorks, and the videos show microphone behavior before and after applying StickShield under typical inaudible and adversarial attacks. Required input parameters: none beyond the provided files.

A.4 Evaluation Workflow

The following experiments describe the operational steps relevant to the paper's key results. The current Artifact Available package contains design files and demonstration videos only. It does not include raw measurements, attack samples, calibration records, repetition counts, or analysis scripts, and it does not claim Functional, Reusable, or Results Reproduced. In E1–E5, "Record ..." lists suggested metadata for independent replication; these records are not part of the current package.

A.4.1 Major claims.

(C1): After applying StickShield, the success rate of attack commands drops significantly, with an attack failure rate above 92%. The quantitative result is reported in Section 6 of the main paper. The current artifact provides qualitative `.mp4` demonstrations for typical inaudible and adversarial attacks; it does not allow independent reproduction of the numerical attack failure rate.

(C2): The recognition accuracy for normal voice commands remains above 95% after applying StickShield. The quantitative result is reported in Section 6 of the main paper. The current artifact does not include the normal-voice measurement data or scripts needed to independently reproduce this value.

(C3): StickShield defends against adversarial audio, inaudible ultrasonic, and laser injection attacks by absorbing acoustic energy in specific frequency bands while preserving the intelligibility of normal voice commands. This claim is supported by the defense and usability experiments in Section 6 of the main paper. Laser injection is discussed in the paper but is not covered by the current artifact package.

A.4.2 Experiments.

(E1): Fabrication and attachment [2–4 machine-hours for printing + 30 person-minutes for attachment and testing].
Preparation: Obtain the `.SLDPRT`/`.STL` files, adjust dimensions if needed, slice the model, and prepare silicone rubber and soft polyurethane rubber according to the README.
Execution: Print the top membrane and bottom cavity structure, remove supports, clean the surface, and attach the patch to the microphone surface with the top side facing outward.
Results: A fabricated StickShield patch is correctly attached to the target microphone. Record printing parameters, material batch, adhesive placement, and photos of the mounted patch.

(E2): Adversarial audio attack test [30 person-minutes; printing time shared with E1].
Preparation: Attach StickShield to the microphone, configure the standard attack device, and prepare synthetic voice commands. Record baseline behavior without StickShield.
Execution: Launch adversarial audio commands into the microphone and observe whether the voice assistant executes the attack commands.
Results: Compare attack success rate with and without StickShield. Expected result: attack failure rate above 92%. Record raw audio, device responses, repetitions, and success/failure labels.

(E3): Inaudible ultrasonic attack test [30 person-minutes; printing time shared with E1].
Preparation: Attach StickShield, configure the ultrasonic attack device, and prepare synthetic commands.
Execution: Inject inaudible ultrasonic commands into the microphone and observe voice assistant behavior.
Results: Compare attack success rate with and without StickShield. Expected result: attack failure rate above 92%. Record signal levels, device responses, repetitions, and success/failure labels.

(E4): Laser injection test (not included in the current artifact package).

Status: This experiment is not covered by the current artifact. The package contains no laser-injection video, raw data, or protocol.
Optional independent replication: Users who independently replicate the laser test must set up the equipment according to the main paper and applicable laser safety procedures, attach StickShield to the microphone surface, perform laser injection with and without StickShield, and compare whether the voice assistant executes injected commands.
Results: Not evaluated in this artifact. For independent replication, compare attack success rate with and without StickShield and record laser parameters, alignment, repetitions, and safety checks.

(E5): Normal voice command recognition test [30 person-minutes; printing time shared with E1].
Preparation: Configure the microphone/voice assistant and prepare a set of normal synthetic voice commands. Record baseline recognition without StickShield.
Execution: Repeat the normal voice commands with StickShield attached and record recognition outcomes.
Results: Compare recognition accuracy with and without StickShield. Expected result: normal command recognition accuracy remains above 95%. Record commands, repetitions, recognition outcomes, and audio conditions.

A.5 Notes on Reusability

Users may adjust the geometric parameters in the `.SLDPRT` files to fit different microphone sizes, copy and paste `.STL` models to construct larger patches covering multiple microphone arrays, or try alternative flexible materials when printing conditions are limited. Material changes may affect acoustic performance, so users should re-characterize the defense and usability performance after substitution. The artifact is intended solely for defensive security research.

A.6 Version

Artifact version 1.0 (November 2026). Based on the LaTeX template for Artifact Evaluation V20220926.